

Semester-wise Pattern for Students Admitted to M. Sc. Physics Programme									
Year	First Semester			U	Second Semester			U	
I	BITS	F103	Introduction to Engineering Design and Prototyping	4	BITS	F111	Thermodynamics	3	
	BIO	F101	Introduction to Biological Sciences	3	BITS	F112	Technical Report Writing	2	
	CHEM	F101	Fundamentals of Chemistry	3	CS	F111	Computer Programming	4	
	MATH	F101	Multivariable Calculus	3	MATH	F113	Probability and Statistics	3	
	PHY	F101	Oscillations and Waves	3	MATH	F102	Linear Algebra and Complex Variables	3	
	BITS	F101	Social Conduct	1	EEE	F111	Electrical Sciences	3	
	BITS	K101	Physical Well-being and Creativity	1*	BITS	F102	Innovation and Design Thinking	1	
				17				19	
II	MATH	F211	Mathematics III	3	ECON	F211	Principles of Economics	3	
			Humanities Electives	3(min)			or	or	
	PHY	F211	Classical Mechanics	4	MGTS	F211	Principles of Management	3	
	PHY	F212	Electromagnetic Theory I	3			Humanities Electives	3(min)	
	PHY	F213	Optics	3	PHY	F241	Electromagnetic Theory II	4	
	PHY	F214	Electricity, Magnetism & Optics Laboratory	2	PHY	F242	Quantum Mechanics I	3	
					PHY	F243	Mathematical Methods of Physics	3	
	BITS	F225	Environmental Studies	3	PHY	F244	Modern Physics Laboratory	2	
			21(min)				18(min)		
Summer BITS F221 Practice School – I (for PS Option Only)									
III			Open/Humanities Electives	3 to 6			Open/Humanities Electives	3 to 6	
	PHY	F311	Quantum Mechanics II	3	PHY	F341	Solid State Physics	3	
	PHY	F312	Statistical Mechanics	3	PHY	F342	Atomic & Molecular Physics	3	
	PHY	F313	Computational Physics	3	PHY	F343	Nuclear & Particle Physics	3	
			Discipline Electives	6(min)	PHY	F344	Advanced Physics Laboratory	3	
							Discipline Electives	3(min)	
			18/21				18/21		
IV			Open Electives	5 to 11	BITS	F412	Practice School-II	20	
			Discipline Electives	6(min)			or	or	
					BITS	F421 T	Thesis	16	
					or	F425 T	Thesis	18-20	
					or	F424 T	Thesis (9) and Electives	15-18	
							(6 to 9)		
			11/17				15/20		

*Discipline Core - 45 Units (15 Courses)

*Discipline Electives - 15 Units (min)-4 Courses(min)